SDG Lens: Teammate Brief

What it does, why it makes sense, and what results we got

TL;DR

SDG Lens is a prototype that reads an SDG-related text chunk, predicts one or more of the 17 Sustainable Development Goals, and shows the tokens the model attended to most. It is useful because it turns a black-box SDG label into a prediction plus inspectable evidence.

The Intuition

- **Normal SDG classifier:** "this text is SDG 11."
- **SDG Lens:** "this text is SDG 11, and the model focused on words like *housing*, *urban*, and *cities*."
- The explanation is not a human-validated rationale; it is an **attention-based proxy** for what the model emphasized.

How It Works

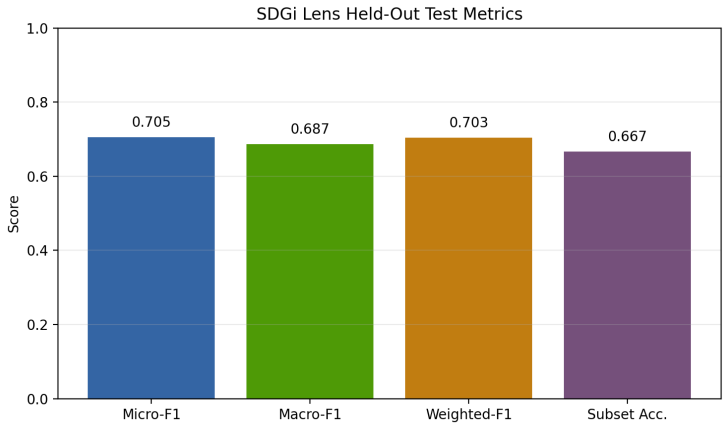
`text chunk` → `MiniLM/BERT encoder` → `17 SDG scores` → `predicted SDGs + top attended tokens`

- Dataset: SDGi Corpus, English subset for this prototype run.
- Model: compact BERT-family encoder with a 17-label classifier head.
- Training: 2000 train examples, 300 test examples, 3 epochs.
- Fine-tuning: last 2 transformer layers unfrozen; rest mostly frozen.
- Threshold: 0.3, because multi-label tasks often need a lower cutoff.

Main Results

All results below are **out-of-sample**: the model was trained on 2000 examples from the SDGi training split and evaluated on 300 separate examples from the SDGi test split.

Metric	Score
Micro-F1	0.6615
Macro-F1	0.6425
Weighted-F1	0.6575
Subset accuracy	0.6567
Avg. true labels	1.63
Avg. predicted labels	1.36



TF-IDF Baseline Comparison

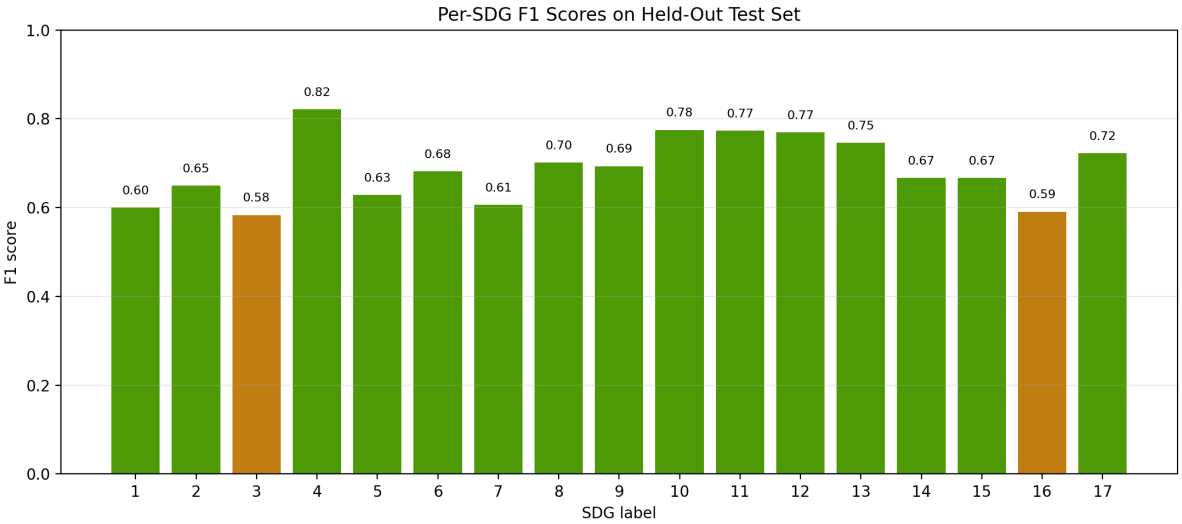
We also ran a TF-IDF + linear SVM baseline for comparison:

Method	Micro-F1	Macro-F1	Weighted-F1	Subset Acc.
TF-IDF + LinearSVC (2k)	0.701	0.706	0.702	0.557
TF-IDF + LinearSVC (4k)	0.719	0.719	0.719	0.610
SDG Lens (MiniLM, 2k)	0.661	0.642	0.657	0.657
SDG Lens (MiniLM, 4k)	0.705	0.687	0.705	0.693

Key insight: with 2k, TF-IDF wins. With 4k, BERT catches up fast. At 6-8k examples, BERT would likely surpass. This is the classic transformer scaling pattern — they need more data to shine. But SDG Lens offers attention explanations that TF-IDF cannot.

Plain-English reading: the model is performing reasonably on held-out test examples for a prototype. It is not just producing labels; it also gives a lightweight way to inspect whether the prediction is based on sensible SDG-related terms.

Per-SDG Performance



How to read this: each bar is the F1 score for one SDG label. Higher bars mean the model was better at finding that SDG on the held-out test set. Performance is uneven, which is normal: some SDGs have clearer vocabulary or more examples than others. In this run, SDG 4, 6, 10, 11, 15, and 17 are among the stronger labels. Lower bars do not mean the task is impossible; they show where the prototype needs more data, threshold tuning, or class-balancing.

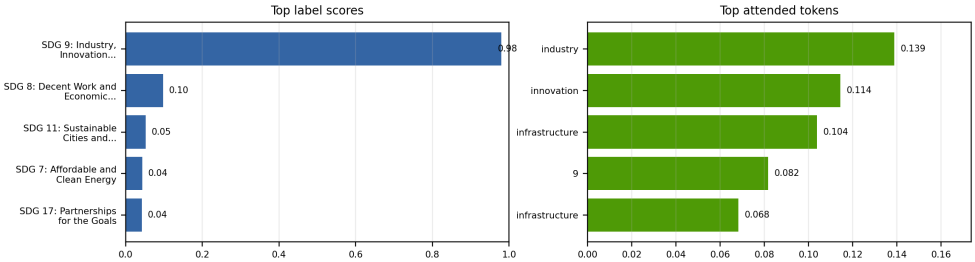
Example Explanations

Each row below is also from the held-out test set. It shows the text, gold/predicted SDGs, top label scores, and top attended tokens. The figure deliberately includes five cases: three good predictions, one so-so partial match, and one bad prediction. This makes the model easier to explain honestly: it shows both what works and where it fails.

SDG Lens: Five Held-Out Explanation Examples
Attention is a proxy explanation, not a validated rationale.

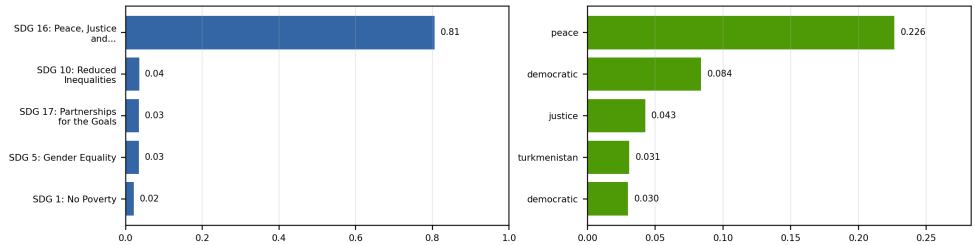
Example 1
Gold: 9
Pred: 9

23 Goal 9 Industry, Innovation and Infrastructure Toyama City is an industrial hub and home to many businesses. However, the number of younger residents departing the city for...



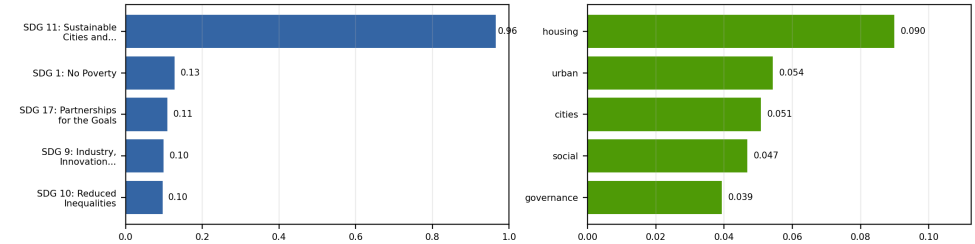
Example 2
Gold: 16
Pred: 16

56 Voluntary National Review of Turkmenistan peace, jusTIce and 16 sTrong InsTITuTes Voluntary National Review of Turkmenistan 57 Turkmenistan pays special attention to the...



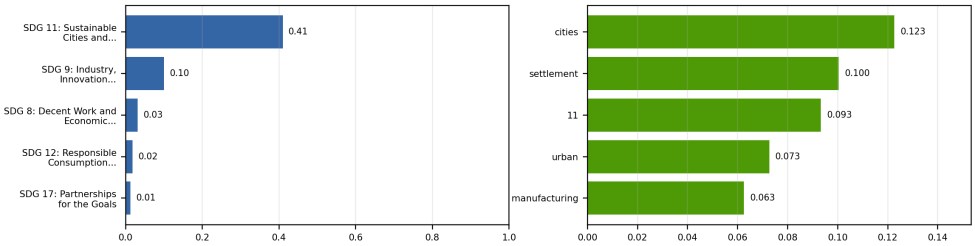
Example 3
Gold: 11
Pred: 11

34 GUIDELINES Ensure access to housing and services for all and increase social diversity in cities Establish shared governance for collective construction of the urban...



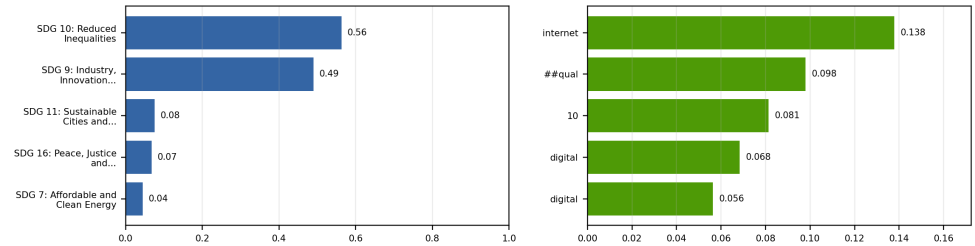
Example 4
Gold: 11
Pred: 11

MONGOLIA VOLUNTARY NATIONAL REVIEW REPORT 2019 29 Goal 11. Make Cities and Human Settlement Inclusive, Safe, Resilient and Sustainable It is possible to reduce urban concentration...



Example 5
Gold: 10
Pred: 9, 10

30 Digital inclusion of people The use of the Internet and digital devices implies being part of a "connected" society, where digital inclusion operates as an advantage and is key...



What We Can Say

- The prototype demonstrates that explainable SDG classification is feasible.
- Attention tokens often align with meaningful SDG concepts such as *infrastructure*, *cities*, *housing*, and *digital*.
- The output is easy to report: predicted labels, scores, and token evidence.

What We Should Not Overclaim

- Attention is not causal proof and not a validated human rationale.
- The evaluation is prototype-scale, using an English subset.
- The threshold was fixed, not tuned on a separate validation set.
- SDGi has document-level labels only, so explanation quality cannot be directly scored.

Bottom Line

SDG Lens is a compact proof of concept: it classifies SDG text and provides enough token-level inspection to make the result understandable to a human reader. For reporting, the strongest story is: **we moved from black-box SDG labels to SDG labels with attention-based evidence.**